

Experiment 3

Gas Leakage Detection System

Aim:

To design a gas leakage detection system using an MQ-5 gas sensor and Arduino, which alerts users through an LED when a gas leak is detected.

Project Description: Gas Leakage Detection System

Introduction:

Gas leakage is a critical safety issue in residential, industrial, and commercial areas. Undetected gas leaks can lead to hazardous situations such as fire explosions, poisoning, and suffocation. Traditional gas leak detection methods rely on human senses, which are not always reliable. This project aims to develop an automated gas leakage detection system using an MQ-5 gas sensor and Arduino. The system provides an early warning by lighting up an LED when it detects flammable gases such as LPG, methane, and natural gas. This ensures quick action to prevent potential hazards.

Objectives:

- To design a low-cost and efficient gas leakage detection system.
- To ensure safety by providing an early warning signal when gas is detected.
- To utilize Arduino Uno as the microcontroller for real-time processing of sensor data.
- To use an MQ-5 gas sensor for detecting combustible gases.
- To implement a LED-based alert system for gas leakage indication

Working Principle:

The system is based on the MQ-5 gas sensor, which is highly sensitive to LPG, methane (CH₄), and natural gas. The sensor operates by detecting changes in gas concentration in the air. When a gas leak occurs, the sensor changes its resistance, which is then converted into an electrical signal. The Arduino reads this signal and

checks if the gas concentration exceeds a preset threshold. If the gas level is higher than the safe limit, the Arduino triggers the LED to glow, signaling the presence of a leak.

Step-by-step process:

1. The MQ-5 sensor continuously monitors the air quality.
2. If the sensor detects gas leakage, it sends an analog signal to the Arduino Uno.
3. The Arduino processes the signal and compares it with a predefined safety threshold.
4. If the gas level exceeds the threshold, the LED turns on, warning of a possible leak.
5. If no gas is detected, the LED remains off.

Components Used:

1. Arduino Uno – The microcontroller that processes sensor data and controls the LED alert.
2. MQ-5 Gas Sensor – Detects gases like LPG, methane, and natural gas.
3. LED – Serves as a visual indicator when gas is detected.
4. Breadboard – Used for making circuit connections without soldering.
5. Jumper Wires – Connects components in the circuit.
6. Resistor – Controls the current flow to prevent damage to components.

Applications:

This gas leakage detection system can be implemented in various fields to enhance safety and prevent hazardous accidents:

- Home Safety: Alerts users about gas leaks in kitchens and prevents explosions.

- Industrial Use: Useful in chemical plants, gas refineries, and factories handling flammable gases.
- Commercial Applications: Can be used in hotels, restaurants, and shopping malls where gas appliances are frequently used.
- Smart Safety Systems: Can be further improved by integrating buzzers, alarms, or IoT-based remote monitoring.